

Smart Delivery Operations Analytics System

Project Report: Business Insights & Operational Performance

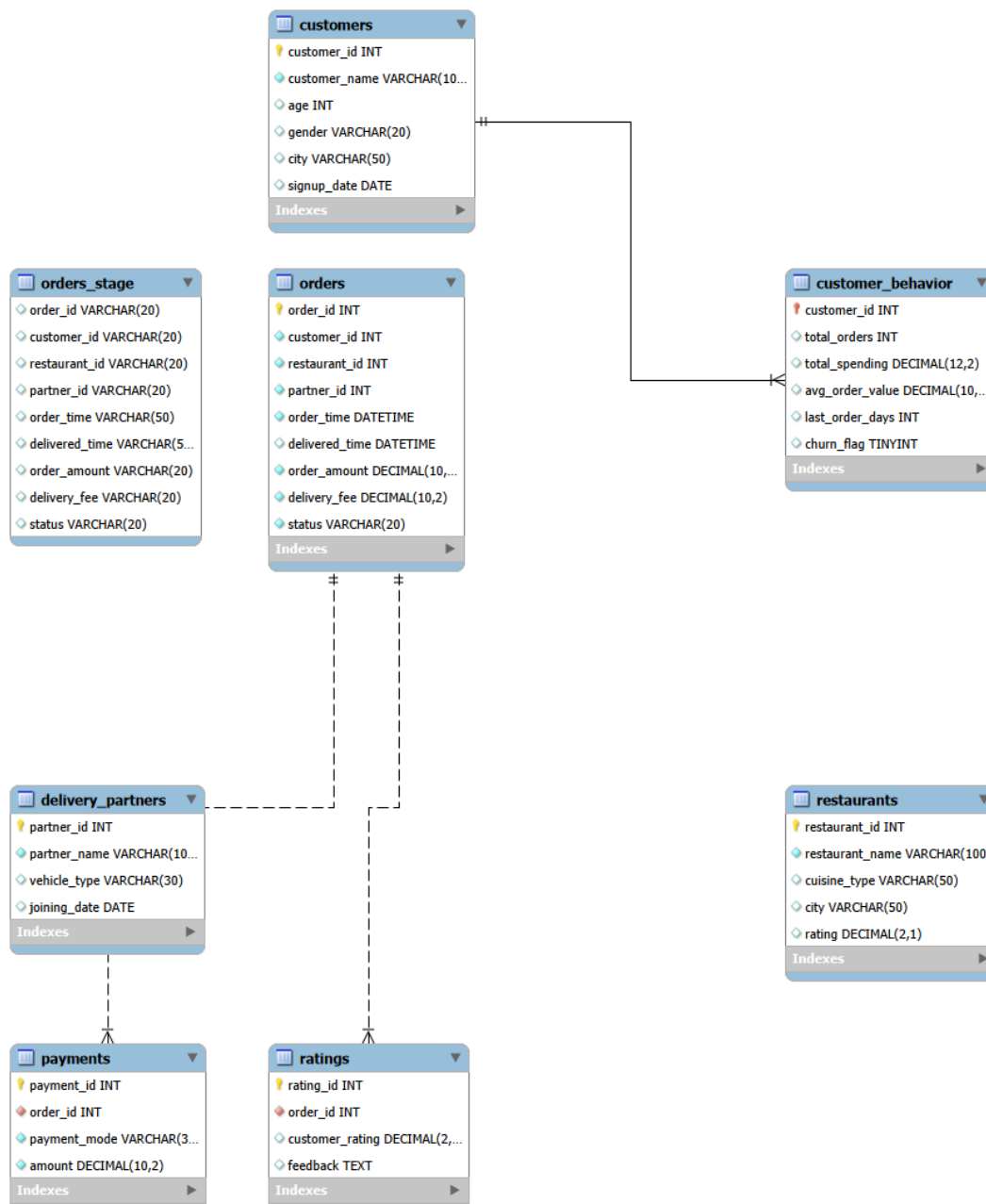
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Project Overview:

I worked on this project to study the day-to-day data from a food delivery platform. My main goal was to find clear ways to make deliveries faster, lower order cancellations, increase revenue, keep customers happy, and track how well drivers are doing.

The dataset included information on customers, restaurants, orders, payments, ratings, and delivery partners. Using SQL, I cleaned and checked all the records to make sure the data was correct. I wrote over 40 different queries to answer important business questions and pull out the insights for this report.

ER Diagram:



Problem 1: Delivery Delays

Key Findings:

During my analysis, I noticed that a few specific restaurants take much longer to deliver food than the rest of the network. Bombay Hub stood out with the worst average delivery time, taking around 62 minutes. I also found that several restaurants under the Royal brand are among the slowest performers on the platform.

When looking at the delivery partners, there is a clear gap in speed. The slowest drivers take more than 60 minutes per delivery on average, while the fastest ones finish their trips in under 50 minutes. This big difference shows that we have room to tighten up operations.

Interestingly, looking at things at a city level did not show massive differences. Hyderabad is the slowest city on average, while Mumbai is the fastest. Looking at the month-on-month trends, delivery speeds stayed mostly flat and stable throughout the reporting period.

Root Causes:

The main reasons for these delays come down to operational issues at specific restaurants and the uneven performance of delivery partners. The data shows that geographic location or which city they are in actually has a very limited impact on delivery speed.

Recommendations:

- Run operational reviews for the ten slowest restaurants on the platform.
- Start monitoring the performance of delivery partners who consistently hit high delivery times.

Estimated Business Impact:

Shaving just five minutes off average delivery time will go a long way. It will boost customer satisfaction, bring in more repeat orders, and lower the number of cancelled orders. Faster deliveries will also naturally lift customer ratings and build better brand loyalty.

Problem 2: Order Cancellations

Key Findings:

My analysis of cancellations shows that a small cluster of restaurants is responsible for a huge chunk of all cancelled orders. Specifically, Classic Point, Royal Corner, Urban Kitchen, and Royal Delight logged the highest numbers of cancellations.

At the city level, Ahmedabad had the highest cancellation rate at 13.49%, with Delhi and Kolkata following closely behind. While the gap between cities is not massive, certain regions clearly struggle with higher cancellation rates more than others.

When I compared delivery times against cancellation volumes, I found a moderate link between the two. Many restaurants with slow delivery times also see more cancellations. However, this isn't the whole story, some restaurants suffer from high cancellations even though their delivery speeds are completely average. This tells me other issues are at play.

Finally, looking at the weekly data, Mondays and Saturdays see the highest number of cancellations, pointing to specific days where operations seem to hit a wall.

Root Causes:

Cancellations happen for a mix of reasons. Slow deliveries are definitely a part of it, especially for slow restaurants. But for the average performing restaurants, the issue might be kitchen delays, running out of food items, or just getting crushed during rush hour.

Recommendations:

- Check what is going wrong at the top 10 restaurants with the most cancellations.
- Watch the restaurants that are both slow and have high cancellations because fixing these will give the biggest win.
- Add extra staff or support on Mondays and Saturdays.
- Set up automatic alerts when a restaurant or city crosses a safe cancellation limit.

Estimated Business Impact:

Lowering the cancellation rate by just 2% to 3% will save a lot of lost money and stop food from being wasted. It also keeps customers happy so they don't leave the app for good.

Problem 3: Revenue Optimisation

Key Findings:

A few star restaurants bring in most of the money. Taste Kitchen made the highest total revenue, with Star Junction and Golden Express right behind them.

City wise, Chennai and Mumbai are our biggest money makers by far. On the other hand, Ahmedabad and Bangalore bring in the least, which means we lack demand or don't have enough good restaurants there.

People love digital payments. UPI is the number one choice for users and brings in the most cash value. Cash on Delivery (COD) is the least popular and brings in the lowest revenue.

Also, revenue goes up and down depending on the month. October 2024 was amazing with a huge jump in sales, but August 2024 saw the biggest drop.

Root Causes:

The revenue is concentrated because certain restaurants are just highly popular or run better. Chennai and Mumbai have a bigger market for us. UPI is booming because customers prefer online payments nowadays. The monthly ups and downs are likely due to seasons, holidays, or specific discount offers.

Recommendations:

- Study what the highest earning restaurants are doing right in terms of operations and marketing, and copy those tactics across other locations.
- Spend more on marketing and sign up more partners in Chennai and Mumbai.
- Create special plans to grow our business in low performing cities like Ahmedabad and Bangalore.
- Keep pushing UPI and digital payments with cashback and discounts.
- Watch monthly revenue trends closely and launch marketing campaigns during historically slow months to smooth out seasonal drops.

Estimated Business Impact:

Helping mid-tier restaurants grow and scaling up business in weaker cities will give a major boost to our total company revenue. Pushing more digital payments will make transactions faster and more convenient for users. Plus, balancing out the slow months will make business growth much more predictable.

Problem 4: Customer Retention

Key Findings:

Looking into customer spending behaviour, I found a small group of highly valuable users who spend significantly more than the average customer. Users like Kiran Shah and Amit Verma have logged the highest lifetime spend on the platform.

The data also shows that a few people order very frequently. They keep the business running and are highly active on the platform.

However, the inactivity data shows a worrying trend: 350 customers have not placed a single order in at least 90 days. This is a large group of people who may have abandoned the platform completely.

The biggest warning sign for losing a customer is when the gap between their orders starts growing.

Root Causes:

Customers leave because they slowly lose interest. If they don't order for a long time, they eventually stop coming back. This usually happens because we aren't sending them regular reminders, good discounts, or loyalty rewards.

Recommendations:

- Run special ads and offers to win back the 350 customers who haven't ordered in 90 days.
- Give personalised rewards and VIP offers to our top spenders so they stay loyal.
- Catch "At Risk" customers who are starting to slow down and give them an incentive before they stop entirely.
- Set up automatic reminder messages at the 60, 75, and 90 day marks of inactivity.

Estimated Business Impact:

It is much cheaper to keep an old customer than to find a new one. Getting even a few inactive users back means easy revenue and less money spent on marketing. Keeping top spenders happy ensures steady earnings for the long run.

Problem 5: Delivery Partner Performance

Key Findings:

Riders show big differences in speed, the number of orders they handle, cancellations, and ratings.

Some riders are superfast, while others take a long time. This comes down to how well they know the roads, how their routes are planned, or how much work they are given.

A few hard-working riders handle a massive chunk of daily deliveries. They are the backbone of the operations.

On the other side, some riders have way too many cancellations. Sriram Pillai has the most, followed by Deepak Patel and Harish Pillai. This needs a proper look, even if it is not entirely their fault.

Root Causes:

Delivery quality is not consistent. The big gaps in speed, ratings, and cancellations show that some riders need help or better management.

High cancellations might happen because of long restaurant delays, bad order assignments, or riders feeling overworked. Poor ratings usually happen when a rider arrives late or behaves poorly.

Recommendations:

- Reward our best riders with bonus programs to keep them motivated.
- Talk to riders who have high cancellation rates or bad ratings to find out what is wrong.
- Give extra training and support to riders who are struggling.

Estimated Business Impact:

Fixing rider issues will directly make customers happier, meaning they will stick with us and trust our brand. Cutting down rider cancellations lowers costs and makes our service reliable. Supporting underperforming riders helps us deliver a good, consistent experience every single time.